RES POLYTECHNIC, BANGALORE-91

CIE Theory Test I

Program	Computer Science and Engineering		Semester- III
Course Name	Data Structure with Python		Test I
Course Code	25CS33I	Duration- 90 min	Marks-50
Name of the Course Coordinator: Vinay Kumar			
Note: Answer any one full question from each section. Each full question carries equal marks.			

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	**a)** Differentiate between a list and a tuple in Python. Give a real-life scenario where you would prefer a tuple over a list. **(5 marks)** **b)** Write a Python function that takes a list of integers as input and returns a new list where: **(10 marks)** • Every even number is replaced by its cube. • Every odd number is increased by 5. **Example** Input: [2, 3, 4, 7, 8, 9] Output: [8, 8, 64, 12, 512, 14] **c)** Write a Python function that takes a list of tuples, where each tuple contains the name of an item and its price. The function should return the name of the most expensive item. **(10 marks)** **Example** Input: [('Pen', 20), ('Book', 150), ('Bag', 600), ('Ruler', 15)] Output: 'Bag'	L2, L3	1	25
	OR			
2	**a)** Explain the common operations that can be performed on a set in Python. Show with examples how to find the union, intersection and difference of two sets. **(5 marks)** **b)** Write a Python function that takes a list of tuples, where each tuple contains the name of a student and the marks scored. The function should return a list of names of the students who scored above the class average. **(10 marks)** **Example** Input: [('Ravi', 72), ('Sita', 85), ('Arun', 60), ('Meena', 91)] Output: ['Sita', 'Meena'] **c)** Write a Python function that takes two sets and returns a tuple containing: **(10 marks)** • The elements common to both sets. • The elements present in either set but not in both. • The elements present in the second set but not in the first set. **Example** Input: set1 = {10, 20, 30, 40} set2 = {30, 40, 50, 60, 70} Output: ({30, 40}, {10, 20, 50, 60, 70}, {50, 60, 70})	L2, L3	1	25

Q. No	Questions	CL	CO	Marks
	SECTION II			

3	Analyze the code and write the output.
	`def sum_nested_elements(nested_structure):`
	`  total_sum = 0`
	`  # Initialize the stack with the outer structure`
	`  stack = [nested_structure]`
	`  print(stack)`
	`  while stack:`
	`  # Take the last element from the stack`
	`  current = stack.pop()`
	`  print(current)`
	`  for item in current:`
	`  # If it's a list or tuple, add to the stack`
	`  if isinstance(item, (list, tuple)):`
	`  print(item)`
	`  stack.append(item)`
	`  print(stack)`
	`  # If it's a number, add to the total`
	`  elif isinstance(item, (int, float)):`
	`  total_sum += item`
	`  return total_sum`
	
	`nested_list = [2, (4, 6), [1, [3, 5]], 7]`
	`total = sum_nested_elements(nested_list)`
	`print(f"Total sum of elements: {total}")`
		L3	2	25

	OR			
4	**a)** Create a Python program to create a collection named **fruits**: Apple, Banana, Mango, Orange, Grapes, Banana, Apple, Guava, Mango, Papaya, Apple, Orange, Banana, Mango, Pineapple, Cherry, Kiwi Compute and display the result with relevant messages for the following: 1. Ask the user to input a fruit name to count how many times it appears in the collection. 2. Ask the user to input a fruit name to find its position in the collection. 3. Get the first 5 fruit names from the collection. 4. Find the unique fruit names from the collection. Suggest the suitable Python concept to solve the above and justify your selection. **b)** *"Dr. A. P. J. Abdul Kalam was born on 15 October 1931 in Rameswaram. He served as the 11th President of India from 2002 to 2007 and worked at ISRO and DRDO."* For the above text, create a Python program to: 1. Measure the length of the text. 2. Count the number of words in the text. 3. Count the occurrence of these stop words – of, the, in, to, and. 4. Convert these words to uppercase and display the text – Kalam, Rameswaram. 5. Find numeric and alphanumeric words from the above text. Suggest the suitable Python concept to solve the above and justify your selection.	L3	2	25

Note for the Course Coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.

Sign of the Course Coordinator Signature of the HOD Signature of the IQAC Chairman